



VIDYARTHI
PROJECT BY
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MANAGEMENT SYSTEM USING PYTHON

- ◆ HOTEL MANAGEMENT SYSTEM
Python Programming Project
- ◆ Presented by: ISHA VARDHAN
26BHI10025, SCHOOL SCAI.
BRANCH ;BTECH CSE IN
HEALTH INFORMATICS
Project: Hotel Management System
Language: Python
- ◆ PYTHON COURSE CODE:-
CSE1021

FEATURES

1. View All Rooms
2. Check Available Rooms Only
3. Book a Room
4. Checkout & Billing
5. Exit

LET SEE THE LOGIC BEHIND THE CODE

1. Data Structure (rooms Dictionary)

Concept: The hotel database uses a Python **Dictionary** where key-value pairs link room numbers to their details.

Format: Room Number -> [Room Type, Per Day Price, Max Capacity, Status / Booking Info]

Dynamic Slot: Index [3] holds "Available" by default. Upon booking, it is replaced with a **nested dictionary** containing guest details:

2. Room Inventory Display (view_all_rooms)

Iteration: Uses a for loop with .items() to traverse every room number and list in the dictionary.

Variable Unpacking: Unpacks values into readable variables (room_type, price, capacity, availability) for clean code structure.

Conditional Logic: Uses an if/else block:

If availability == "Available", it displays the room as free.

Otherwise, it accesses availability["name"] from the nested dictionary to show who currently occupies the room

3. Availability Filter (check_availability)

Counter Pattern: Initializes a tracking variable (count = 0).

Filtering Loop: Loops through the dictionary and checks index [3] of each room.

Logic: If info[3] == "Available", it prints the room details and increments count += 1.

Zero-Check: After completing the loop, an if count == 0: condition alerts the user if all rooms are fully booked

4.Booking & Validation (book_room)

Existence Check: Verifies if room_no in rooms to ensure the room exists.

Availability Guard: Ensures rooms[room_no][3] == "Available" to prevent overwriting existing bookings.

Capacity Validation: Compares user input guest_count against rooms[room_no][2]. If guest_count > max_capacity, the booking is blocked.

Duration Check: Confirms that stay days are greater than 0.

Error Handling: Uses try/except ValueError blocks to catch invalid non-numeric inputs without crashing the application.

5. Checkout & Bill Calculation (checkout_and_billing)

Active Booking Verification: Ensures the requested room is currently occupied before attempting checkout.

Math Logic: Retrieves the price per day and stay duration from memory and calculates the final cost:

$$\text{Total Bill} = \text{Price Per Day} \times \text{Days Stayed}$$

Status Reset: Overwrites index [3] back to "Available", making the room immediately available for future bookings.

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GIVEN BELOW THE FLOW CONTROL OF MY PROJECT

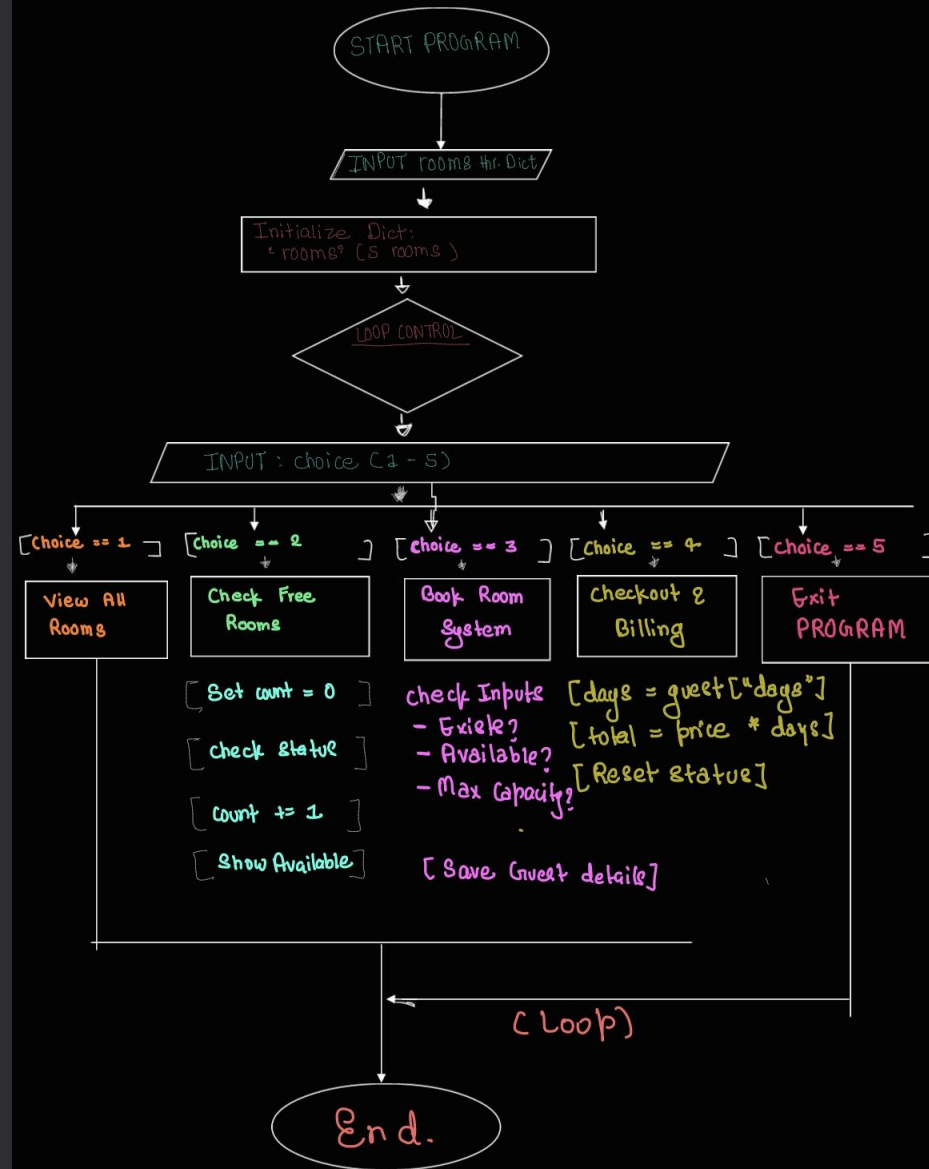
(I HAVE USED HANDMADE LOOP CONTROL FROM MS WORD TO SHOW THE AUTHRNTICITY OF THE PROJECT)



FOCUS

1:12:08

▶



THE UPCOMING PPT ARE OUTPUT THAT I GOT AFYER RUNNING MY CODE

TERMINAL ADDRESS DISCRPTION TESTED ON VS CODE

```
C:\Users\ASUS\Documents\VIDYARTHI PROJECT ISHA26BHI10025> &  
C:\Users\ASUS\AppData\Local\Python\pythoncore-3.14-64\python.exe  
"c:/Users/ASUS/Documents/VIDYARTHI PROJECT ISHA26BHI10025/HOTEL  
MANAGEMENT.PY"
```

Explorer

VIDYARTHI PROJECT ISHA26BHI10025HOTEL MANAGEMENT.PY

OutlineTimeline

Problems Output Debug Console Terminal Ports

```
PS C:\Users\ASUS\Documents\VIDYARTHI PROJECT ISHA26BHI10025> & C:\Users\ASUS\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/ASUS/Documents/VIDYARTHI PROJECT ISHA26BHI10025/HOTEL MANAGEMENT.PY"

=====
HOTEL MANAGEMENT SYSTEM
=====
1. View All Rooms
2. Check Available Rooms Only
3. Book a Room
4. Checkout & Billing
5. Exit
Enter choice (1-5): 1

--- ALL ROOMS STATUS ---
Room 101 | Single Bed | $50/day | Max Guests: 1 -> AVAILABLE
Room 102 | Double Bed | $80/day | Max Guests: 2 -> AVAILABLE
Room 201 | Double Bed | $80/day | Max Guests: 2 -> AVAILABLE
Room 202 | Family Suite | $140/day | Max Guests: 4 -> AVAILABLE
Room 301 | Luxury Suite | $220/day | Max Guests: 5 -> AVAILABLE

=====
HOTEL MANAGEMENT SYSTEM
=====
1. View All Rooms
2. Check Available Rooms Only
3. Book a Room
4. Checkout & Billing
5. Exit
Enter choice (1-5): 2

--- AVAILABLE ROOMS ONLY ---
[x] Room 101 (Single Bed) - $50/day (Max Guests: 1)
[x] Room 102 (Double Bed) - $80/day (Max Guests: 2)
[x] Room 201 (Double Bed) - $80/day (Max Guests: 2)
[x] Room 202 (Family Suite) - $140/day (Max Guests: 4)
[x] Room 301 (Luxury Suite) - $220/day (Max Guests: 5)

Total rooms free: 5
```

Explorer

VIDYARTHI PROJECT ISHA26BHI10025

HOTEL MANAGEMENT.PY

Outline

Timeline

Problems Output Debug Console Terminal Ports

Total rooms free: 5

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HOTEL MANAGEMENT SYSTEM

=====

1. View All Rooms

2. Check Available Rooms Only

3. Book a Room

4. Checkout & Billing

5. Exit

Enter choice (1-5): 3

NEW GUEST BOOKING ---

Enter room number to book: 101

Enter guest count (Max 1): 1

Enter number of stay days: 3

Enter guest full name: ISHA VARDHAN

Enter contact phone number: 8294158115

Success! Room 101 booked for ISHA VARDHAN for 3 day(s).

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HOTEL MANAGEMENT SYSTEM

=====

1. View All Rooms

2. Check Available Rooms Only

3. Book a Room

4. Checkout & Billing

5. Exit

Enter choice (1-5): 4

CHECKOUT & BILLING ---

Enter room number: 101

BILL RECEIPT ---

Guest Name: ISHA VARDHAN

Phone Number: 8294158115

Days Stayed: 3

The image shows a Visual Studio Code (VS Code) editor window with a dark theme. The Explorer sidebar on the left displays a project named 'VIDYARTHI PROJECT ISHA26BHI10025' containing a file 'HOTEL MANAGEMENT.PY'. The main editor area shows the contents of this file, which is a Python script for a hotel management system. The script includes a menu with five options: 1. View All Rooms, 2. Check Available Rooms Only, 3. Book a Room, 4. Checkout & Billing, and 5. Exit. The user has interacted with the program, entering '3' to book a room, '101' for the room number, and '4' for the checkout and billing option. The terminal output shows the program's execution, including a success message for booking a room and a detailed bill receipt for a 3-day stay at \$150. The program has exited, and the terminal shows the command prompt 'PS C:\Users\ASUS\Documents\VIDYARTHI PROJECT ISHA26BHI10025>'. The bottom status bar shows the file is saved and the project is open.

```
Enter number of stay days: 3
Enter guest full name: ISHA VARDHAN
Enter contact phone number: 8294158115

Success! Room 101 booked for ISHA VARDHAN for 3 day(s).

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HOTEL MANAGEMENT SYSTEM
=====
1. View All Rooms
2. Check Available Rooms Only
3. Book a Room
4. Checkout & Billing
5. Exit
Enter choice (1-5): 4

--- CHECKOUT & BILLING ---
Enter room number: 101

--- BILL RECEIPT ---
Guest Name: ISHA VARDHAN
Phone Number: 8294158115
Days Stayed: 3
Price Per Day: $ 50
Total Bill: $ 150
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Checkout complete. Room is free again.

=====
HOTEL MANAGEMENT SYSTEM
=====
1. View All Rooms
2. Check Available Rooms Only
3. Book a Room
4. Checkout & Billing
5. Exit
Enter choice (1-5): 5
Exiting project. Thank you!
PS C:\Users\ASUS\Documents\VIDYARTHI PROJECT ISHA26BHI10025>
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REPORT

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